Yomna Khaled Alshemy

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Yomna Alshemy — Live Systems

Education

Misr University for Science and Technology (MUST) Cairo, Egypt
Bachelor of Science in Computer Science June, 2025
GPA: 3.86/4.00 • Ranked 12th in class
Relevant coursework: Databases, Web Development, Software Engineering, Computer Networks, Object-Oriented Programming, Data Structures and Algorithms, Computer Security, Pattern Recognition.

Experience

Senior FinCrime Analyst — Automation & Investigations Dubai, UAE (Remote)
Ziina August 2025 – Present

- Own end-to-end investigations across fraud, KYB, sanctions/PEP, transaction monitoring, chargebacks, and Emirates NBD partner-bank RFIs; cleared all 233 assigned sanctions/PEP alerts and handled 749 Intercom cases (584 self-initiated).
- Detected and worked 5 organized fraud typologies end-to-end — including a Libyan card-fraud ring spanning 422 cards across 24 BINs — and independently traced a major coordinated card attack behind the company's worst fraud month, mapping 60+ accounts for ban via IBAN/P2P chains.
- Designed and built Sentinel, a self-initiated tool suite (police-report generation, Emirates NBD RFI handling, and a 35,000-character SQL search spanning 42 CTEs and 13 warehouse/staging tables) that cut case response time from 9 to 6.9 minutes despite handling more cases per email, and now runs the team's law-enforcement response process.
- Built three production Python automations for KYB remediation (dry-run safe, resumable, rate-limit-aware) that cut field-set failures from 303 to 1 and reduced the remediation queue from ~1,495 to ~1,167 cases.
- Own Ziina's chargeback defense process end-to-end — researched the full dispute lifecycle and Visa CE3.0 compelling-evidence standard, authored a 5-network reason-code playbook and a 5-document SOP suite, and personally built 111+ representment cases by hand; piloted two self-submitted defenses — both reversed — and filed three formal Amex defense packets.
- Initiated and built a RAG-based chargeback-defense automation: an n8n-orchestrated, two-stage (retrieval/reasoning → generation) LLM pipeline that drafts representment packages straight from the Dispute Register for analyst review, architected to replace a third-party defense vendor.
- Personally dispositioned 512 KYB resubmission cases across four workstreams, the largest individual contribution on three of four, recovering accounts that had sat unresolved for months.
- Self-initiated an audit of ML-bypassing whitelist exemptions (10 accounts, 5,986 transactions, 70.73% average failure rate), publishing a 90-day remediation plan; tooling and write-ups cited by leadership as reference material for the wider team.

Research Assistant: AI-Powered Lupus Diagnosis Assistant Project Cairo, Egypt
Misr University for Science and Technology (MUST) Sept 2024 - June 2025

- Led the development of a mobile app to assist in lupus diagnosis, applying an SVM machine learning model aligned with EULAR/ACR diagnostic criteria.
- Designed and implemented algorithms to assess lupus probability from patient-reported data, facilitating timely screening in low-resource settings.
- Conducted data preprocessing, feature engineering, and model validation to ensure performance reliability.
- Integrated AI outputs into a full-stack mobile application for practical use in healthcare settings.

IT & Compliance Support Intern Cairo, Egypt
Misr University for Science and Technology (MUST) at SCEEP July 2022 – Feb 2025

- Monitored student and staff activity across internal systems to catch unusual access patterns and reported anything suspicious to the IT team.
- Supported compliance reviews of internal digital platforms, contributing to UI/UX audits and suggesting interface improvements to ensure both functionality and security.
- Logged and followed up on access-related issues and system tickets, keeping records up to date and supporting internal case tracking.

Projects

ITQAN – Full-Stack Cambridge IGCSE Revision Platform itqan-web-dy2r.onrender.com

Next.js, NestJS, TypeScript, PostgreSQL, Redis

- Built and shipped a full-stack revision platform (~61K lines of TypeScript, 147 API routes, 1,025 automated tests) serving three role-based portals — students, tutors, parents — with assignments, marking, and progress tracking, used across the UAE and Egypt.
- Engineered a from-scratch cipher-decoding pipeline to recover text from PDFs with corrupted font maps (known-plaintext cribs, a syllabus-trained trigram model, coordinate-ascent solving), reaching 98.1% character accuracy and 100% question-anchoring across 20 real exam papers with zero external services.
- Built privacy-preserving analytics for a product used by minors — daily-rotating salted visitor hashes, server-verified attribution, no IP ever stored.
- Instituted a verification-first engineering discipline: 11 custom audit scripts (contrast, dead-code, contract/DDL consistency) and mutation-tested fixes, catching defects before they shipped.
- Rebuilt the design system from a gradient-heavy theme to a token-driven neutral UI (semantic color tokens, single accent, theme-adaptive SVG favicon), with automated WCAG AA contrast checks across every page and theme that caught unstyled buttons before launch.

CuraCare - Full-Stack Healthcare Platform mini-carepulse.onrender.com

Next.js | Twilio, TypeScript, TailwindCSS

- Built a secure platform with structured data models for patient intake, appointment flow, and medical records. Integrated admin/provider dashboards and implemented SQL-based access controls and audit trails to support performance monitoring.

Lupira - Full-Stack AI-Powered Lupus Detection App lupira-mulo.onrender.com

Flutter, Python, Scikit-learn

- Designed and deployed a mobile app leveraging SVM models for early lupus risk prediction. Engineered custom datasets, mapped clinical inputs to diagnostic probabilities, and visualized output metrics to guide patient self-assessments and clinical follow-ups.

Horizon Banking - Full-Stack Financial Management Platform mini-banking.onrender.com

Next.js, Appwrite, Sentry, WebStorm

- Developed a responsive fintech app with secure user auth, cross-account transfers, and data-driven spending analysis. Built SQL queries and reporting logic to generate user-level insights and budgeting recommendations using anonymized transaction data.

Additional

Professional Skills: Fraud investigation, AML/KYB compliance, sanctions & PEP screening, chargeback dispute resolution, data analysis, SQL, communication, documentation review, problem-solving, case tracking, attention to detail, teamwork, cross-functional collaboration, investigative mindset, adaptability

Language Skills: Arabic (Native), English (Bilingual Proficiency)

Technical Skills: JavaScript, Kali Linux, Python, PHP, Bash, HTML, CSS, TypeScript, Flutter, Node.js, NestJS, Express.js, MongoDB, MySQL, PostgreSQL, TypeORM, Redis, Scikit-learn, Next.js, Appwrite, TailwindCSS, Visual Studio Code, Twilio, Git and GitHub, BurpSuite, WireShark, UI/UX design, design systems, accessibility (WCAG AA)

Certifications

- **Certified Ethical Hacker (CEH Master, 312-50v13)**, EC-Council – *In Progress, targeting Dec 2026*
- **JavaScript Essentials 1**, Cisco Networking Academy – *Dec 2024*
- **JavaScript Essentials 2**, Cisco Networking Academy – *Feb 2025*
- **MongoDB Node.js Developer Path**, MongoDB University – *Feb 2025*
- **Responsive Web Design**, FreeCodeCamp – *Mar 2025*
- **Introduction to Cybersecurity**, Cisco Networking Academy – *Nov 2024*
- **Advent of Cyber**, TryHackMe – *Dec 2024*
- **Object-Oriented Programming**, *Dec 2024*
- **Scripting Languages: PHP, Python, JavaScript, Bash**, Completion – *Dec 2024*